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Bandwidth part configuration for reduced capability devices

Abstract

Methods, systems, and devices for wireless communications are described. A user equipment (UE) may receive a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The UE may select a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity. The UE may transmit, as part of a random access procedure, the random access preamble within the random access occasion.

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Background/Summary

TECHNICAL FIELD
(1) The following relates to wireless communications, including bandwidth part configuration for reduced capability devices.
BACKGROUND
(2) Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).
(3) In some wireless communications system, a reduced capability wireless device may establish communications with a network. However, methods for establishing communications between a reduced capability wireless device and a network may be improved.
SUMMARY
(4) The described techniques relate to improved methods, systems, devices, and apparatuses that support bandwidth part configuration for reduced capability devices. For example, a user equipment (UE) (e.g., a reduced capability device) may receive a synchronization signal block message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity. The synchronization signal block message may also indicate a plurality of random access occasions for the UE to use for transmission of a first random access message to the network entity. The plurality of random access occasions may be in frequency resources that are outside of the first initial uplink bandwidth part. For example, the plurality of random access occasions may be in respective initial uplink bandwidth parts (the first initial uplink bandwidth part and additional initial uplink bandwidth parts). In another example, the plurality of random access occasions may be outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part. The UE may select a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity. The UE may transmit, as part of a random access procedure, the random access preamble within the random access occasion.
(5) A method for wireless communications at a UE is described. The method may include receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part, selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity, and transmitting, as part of a random access procedure, the random access preamble

within the random access occasion.

(6) An apparatus for wireless communications at a UE is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part, select a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity, and transmit, as part of a random access procedure, the random access preamble within the random access occasion.

(7) Another apparatus for wireless communications at a UE is described. The apparatus may include means for receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part, means for selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity, and means for transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

(8) A non-transitory computer-readable medium storing code for wireless communications at a UE is described. The code may include instructions executable by a processor to receive a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part, select a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity, and transmit, as part of a random access procedure, the random access preamble within the random access occasion.

(9) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the message may include operations, features, means, or instructions for receiving an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the set of multiple random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.

(10) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for communicating additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble may be transmitted.

(11) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, incrementing a preamble power ramping parameter based on failure of the transmission of the random access preamble and on whether the first attempted initial uplink bandwidth part may be to be reused for retransmission of the random access preamble, and retransmitting the random access preamble in accordance with the preamble power ramping parameter.

(12) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, incrementing a preamble power ramping parameter based on failure of the transmission of the random access preamble, regardless of whether the first attempted initial uplink bandwidth part may be to be reused for retransmission of the random access preamble, and retransmitting the random access preamble in accordance with the preamble power ramping parameter.
(13) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining that transmission of the random access preamble failed and retransmitting the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, where the retransmitting may be in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.
(14) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts and retransmitting the random access preamble via the first attempted initial uplink bandwidth part, where the retransmitting may be in accordance with a maximum number of retransmissions parameter that applies to retransmissions per initial uplink bandwidth part.
(15) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, selecting the random access occasion may include operations, features, means, or instructions for monitoring an initial downlink bandwidth part for paging occasions during the random access procedure, identifying, from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, a corresponding initial uplink bandwidth part that corresponds to the initial downlink bandwidth part, and selecting the random access occasion that may be within the corresponding initial uplink bandwidth part based on the corresponding initial uplink bandwidth part corresponding to the initial downlink bandwidth part.
(16) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, at most one of the set of multiple the random access occasions may be within the first initial uplink bandwidth part, a remainder of the set of multiple the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.
(17) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving information that may be indicative of at least one of an offset parameter or a binary direction parameter, where the binary direction parameter indicates a direction in which the offset parameter may be to be applied with respect to a baseline resource block and determining one or more of the frequency resources that may be outside of the first initial uplink bandwidth part based on the information.
(18) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the information that may be indicative of at least one of the offset parameter or the binary direction parameter may include operations, features, means, or instructions for receiving indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis.
(19) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, determining one or more of the frequency resources that may be outside of the first initial uplink bandwidth part may include operations, features, means, or instructions for determining, based on the offset parameter, the binary direction parameter, the baseline resource block, and a first frequency resource associated with a first random access occasion of the set of multiple random access occasions and determining additional frequency resources associated with other random access occasions of the set of multiple random access occasions, the additional frequency resources determined in relation to the first frequency resource.
(20) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for communicating additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a selected initial uplink bandwidth part that corresponds to the random access occasion.
(21) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for receiving a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(22) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for transmitting a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(23) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for transmitting a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.
(24) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the message may be received through a system information block, a synchronization signal block, downlink control information, a medium access control control element, radio resource control signaling, a handover command, or any combination thereof.
(25) A method for wireless communications at a network entity is described. The method may include transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part and receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.
(26) An apparatus for wireless communications at a network entity is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to transmit a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part and receive, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.
(27) Another apparatus for wireless communications at a network entity is described. The apparatus

may include means for transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part and means for receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.

(28) A non-transitory computer-readable medium storing code for wireless communications at a network entity is described. The code may include instructions executable by a processor to transmit a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part and receive, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.

(29) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the message may include operations, features, means, or instructions for transmitting an indication of one or more additional initial uplink bandwidth parts for use by a UE in the communications with the network entity, each of the set of multiple random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.

(30) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for communicating additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble may be transmitted.

(31) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining that transmission of the random access preamble failed and receiving a retransmission of the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, where the retransmitting may be in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.

(32) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, at most one of the set of multiple the random access occasions may be within the first initial uplink bandwidth part, a remainder of the set of multiple the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.

(33) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting information that may be indicative of at least one of an offset parameter or a binary direction parameter, where the binary direction parameter indicates a direction in which the offset parameter may be to be applied with respect to a baseline resource block and determining one or more of the frequency resources that may be outside of the first initial uplink bandwidth part based on the information.

(34) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the information that may be indicative of at least one of the offset parameter or the binary direction parameter may include operations, features, means, or instructions for transmitting indications of at least one of the offset parameter or the binary

direction parameter on a per-random access occasion basis.
(35) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for communicating additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a selected initial uplink bandwidth part that corresponds to the random access occasion.
(36) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for transmitting a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(37) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for receiving a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(38) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the additional one or more random access messages may include operations, features, means, or instructions for receiving a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an example of a wireless communications system that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(2) FIGS. **2**A and **2**B illustrate examples of a bandwidth part configuration that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(3) FIG. **3** illustrates an example of a bandwidth part configuration that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(4) FIGS. **4**A, **4**B, and **4**C illustrate examples of a bandwidth part configuration that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(5) FIG. **5** illustrates an example of a process flow that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(6) FIGS. **6** and **7** show block diagrams of devices that support bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(7) FIG. **8** shows a block diagram of a communications manager that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(8) FIG. **9** shows a diagram of a system including a device that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.

(9) FIGS. **10** and **11** show block diagrams of devices that support bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(10) FIG. **12** shows a block diagram of a communications manager that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(11) FIG. **13** shows a diagram of a system including a device that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(12) FIGS. **14** through **17** show flowcharts illustrating methods that support bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(13) Some wireless communications systems may support user equipment (UEs) with different capabilities. For example, a wireless communications system may support higher capability UEs with low latency and high data throughput, such as UEs that support ultra-reliable low latency communications (URLLC). The wireless communications system may also support lower capability UEs, or reduced capability UEs, with reduced peak data throughput, latency, reliability, bandwidth, or other characteristics or capabilities. For example, some devices may operate within a limited radio frequency (RF) bandwidth. However, such bandwidth parameters may limit a number of random access occasions available to reduced capability UEs during a random access procedure and may further limit a number of reduced capability UEs that may be accommodated within an initial bandwidth part used for the random access procedure.
(14) To reduce or eliminate such challenges, a network entity may indicate multiple random access occasions that may be associated with one or more initial uplink bandwidth parts for performing a random access procedure. In some examples, a network entity may configure multiple initial uplink bandwidth parts each including a random access occasion within the corresponding initial uplink bandwidth part. In other examples, a network entity may configure multiple random access occasions associated with a single initial uplink bandwidth part (e.g., through frequency domain multiplexing) and one or more of the multiple random access occasions may lie in a frequency range falling outside of a frequency range of the initial uplink bandwidth part. A reduced capability UE may perform a portion of a random access procedure (such as transmitting a random access preamble) within a random access occasion outside of the initial uplink bandwidth part and may perform one or more other portions of the random access procedure within the initial uplink bandwidth part.
(15) In some examples, a network entity may configure the multiple random access occasions in various ways. For example, the network entity may configure a binary direction parameter that indicates a direction away from a reference point (e.g., higher frequency or lower frequency) in which multiple random access occasions may be allocated. In some examples, the network entity may configure a variable offset or reference point from which the random access occasions may be located by the UE (e.g., instead of using a physical resource block of the bandwidth part as such a reference). Additionally, or alternatively, the network entity may configure an offset parameter for each random access occasion that may indicate its frequency resource allocation.
(16) Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of the disclosure are then described with reference to bandwidth part configurations and a process flow. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to bandwidth part configuration for reduced capability devices.
(17) FIG. **1** illustrates an example of a wireless communications system **100** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The wireless communications system **100** may include one or

more network entities **105**, one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

(18) The network entities **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities **105** and UEs **115** may wirelessly communicate via one or more communication links **125** (e.g., a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (e.g., a geographic coverage area) over which the UEs **115** and the network entity **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

(19) The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** or network entities **105**, as shown in FIG. **1**.

(20) As described herein, a node of the wireless communications system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (e.g., any network entity described herein), a UE **115** (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, or the like may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

(21) In some examples, network entities **105** may communicate with the core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via one or more backhaul communication links **120** (e.g., in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities **105** may communicate with one another over a backhaul communication link **120** (e.g., in accordance with an X2, Xn, or other interface protocol) either directly (e.g., directly between network entities **105**) or indirectly (e.g., via a core network **130**). In some examples, network entities **105** may communicate with one another via a midhaul communication link **162** (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication links **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (e.g., an electrical link, an optical fiber link), one or more wireless links (e.g., a radio

link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** through a communication link **155**.

(22) One or more of the network entities **105** described herein may include or may be referred to as a base station **140** (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity **105** (e.g., a base station **140**) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within a single network entity **105** (e.g., a single RAN node, such as a base station **140**).

(23) In some examples, a network entity **105** may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among two or more network entities **105**, such as an integrated access backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU) **160**, a distributed unit (DU) **165**, a radio unit (RU) **170**, a RAN Intelligent Controller (RIC) **175** (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) **180** system, or any combination thereof. An RU **170** may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (e.g., separate physical locations). In some examples, one or more network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

(24) The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending upon which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, and any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (e.g., layer 3 (L3), layer 2 (L2)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaption protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** may be connected to one or more DUs **165** or RUs **170**, and the one or more DUs **165** or RUs **170** may host lower protocol layers, such as layer 1 (L1) (e.g., physical (PHY) layer) or L2 (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (e.g., via one or more RUs **170**). In some cases, a functional split between a CU **160** and a DU **165**, or between a DU **165** and an RU **170** may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to one or more DUs **165** via a midhaul communication link **162** (e.g., F1, F1-c, F1-u), and a DU **165** may be connected to one or more

RUs **170** via a fronthaul communication link **168** (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities **105** that are in communication over such communication links.

(25) In wireless communications systems (e.g., wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network **130**). In some cases, in an IAB network, one or more network entities **105** (e.g., IAB nodes **104**) may be partially controlled by each other. One or more IAB nodes **104** may be referred to as a donor entity or an IAB donor. One or more DUs **165** or one or more RUs **170** may be partially controlled by one or more CUs **160** associated with a donor network entity **105** (e.g., a donor base station **140**). The one or more donor network entities **105** (e.g., IAB donors) may be in communication with one or more additional network entities **105** (e.g., IAB nodes **104**) via supported access and backhaul links (e.g., backhaul communication links **120**). IAB nodes **104** may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by DUs **165** of a coupled IAB donor. An IAB-MT may include an independent set of antennas for relay of communications with UEs **115**, or may share the same antennas (e.g., of an RU **170**) of an IAB node **104** used for access via the DU **165** of the IAB node **104** (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB nodes **104** may include DUs **165** that support communication links with additional entities (e.g., IAB nodes **104**, UEs **115**) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., one or more IAB nodes **104** or components of IAB nodes **104**) may be configured to operate according to the techniques described herein.

(26) For instance, an access network (AN) or RAN may include communications between access nodes (e.g., an IAB donor), IAB nodes **104**, and one or more UEs **115**. The IAB donor may facilitate connection between the core network **130** and the AN (e.g., via a wired or wireless connection to the core network **130**). That is, an IAB donor may refer to a RAN node with a wired or wireless connection to core network **130**. The IAB donor may include a CU **160** and at least one DU **165** (e.g., and RU **170**), in which case the CU **160** may communicate with the core network **130** over an interface (e.g., a backhaul link). IAB donor and IAB nodes **104** may communicate over an F1 interface according to a protocol that defines signaling messages (e.g., an F1 AP protocol). Additionally, or alternatively, the CU **160** may communicate with the core network over an interface, which may be an example of a portion of backhaul link, and may communicate with other CUs **160** (e.g., a CU **160** associated with an alternative IAB donor) over an Xn-C interface, which may be an example of a portion of a backhaul link.

(27) An IAB node **104** may refer to a RAN node that provides IAB functionality (e.g., access for UEs **115**, wireless self-backhauling capabilities). A DU **165** may act as a distributed scheduling node towards child nodes associated with the IAB node **104**, and the IAB-MT may act as a scheduled node towards parent nodes associated with the IAB node **104**. That is, an IAB donor may be referred to as a parent node in communication with one or more child nodes (e.g., an IAB donor may relay transmissions for UEs through one or more other IAB nodes **104**). Additionally, or alternatively, an IAB node **104** may also be referred to as a parent node or a child node to other IAB nodes **104**, depending on the relay chain or configuration of the AN. Therefore, the IAB-MT entity of IAB nodes **104** may provide a Uu interface for a child IAB node **104** to receive signaling from a parent IAB node **104**, and the DU interface (e.g., DUs **165**) may provide a Uu interface for a parent IAB node **104** to signal to a child IAB node **104** or UE **115**.

(28) For example, IAB node **104** may be referred to as a parent node that supports communications for a child IAB node, and referred to as a child IAB node associated with an IAB donor. The IAB donor may include a CU **160** with a wired or wireless connection (e.g., a backhaul communication

link **120**) to the core network **130** and may act as parent node to IAB nodes **104**. For example, the DU **165** of IAB donor may relay transmissions to UEs **115** through IAB nodes **104**, and may directly signal transmissions to a UE **115**. The CU **160** of IAB donor may signal communication link establishment via an F1 interface to IAB nodes **104**, and the IAB nodes **104** may schedule transmissions (e.g., transmissions to the UEs **115** relayed from the IAB donor) through the DUs **165**. That is, data may be relayed to and from IAB nodes **104** via signaling over an NR Uu interface to MT of the IAB node **104**. Communications with IAB node **104** may be scheduled by a DU **165** of IAB donor and communications with IAB node **104** may be scheduled by DU **165** of IAB node **104**.
(29) In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support bandwidth part configuration for reduced capability devices as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (e.g., a base station **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., IAB nodes **104**, DUs **165**, CUs **160**, RUs **170**, RIC **175**, SMO **180**).
(30) A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a multimedia/entertainment device (e.g., a radio, a MP3 player, or a video device), a camera, a gaming device, a navigation/positioning device (e.g., GNSS (global navigation satellite system) devices based on, for example, GPS (global positioning system), Beidou, GLONASS, or Galileo, or a terrestrial-based device), a tablet computer, a laptop computer, a netbook, a smartbook, a personal computer, a smart device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, virtual reality goggles, a smart wristband, smart jewelry (e.g., a smart ring, a smart bracelet)), a drone, a robot/robotic device, a vehicle, a vehicular device, a meter (e.g., parking meter, electric meter, gas meter, water meter), a monitor, a gas pump, an appliance (e.g., kitchen appliance, washing machine, dryer), a location tag, a medical/healthcare device, an implant, a sensor/actuator, a display, or any other suitable device configured to communicate via a wireless or wired medium. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.
(31) The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.
(32) The UEs **115** and the network entities **105** may wirelessly communicate with one another via one or more communication links **125** (e.g., an access link) over one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a RF spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (e.g., LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component

carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (e.g., a base station **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (e.g., directly or via one or more other network entities **105**).

(33) In some examples, such as in a carrier aggregation configuration, a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute RF channel number (EARFCN)) and may be positioned according to a channel raster for discovery by the UEs **115**. A carrier may be operated in a standalone mode, in which case initial acquisition and connection may be conducted by the UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode, in which case a connection is anchored using a different carrier (e.g., of the same or a different radio access technology).

(34) The communication links **125** shown in the wireless communications system **100** may include downlink transmissions (e.g., forward link transmissions) from a network entity **105** to a UE **115**, uplink transmissions (e.g., return link transmissions) from a UE **115** to a network entity **105**, or both, among other configurations of transmissions. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications (e.g., in a TDD mode).

(35) A carrier may be associated with a particular bandwidth of the RF spectrum and, in some examples, the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a set of bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (e.g., the network entities **105**, the UEs **115**, or both) may have hardware configurations that support communications over a particular carrier bandwidth or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include network entities **105** or UEs **115** that support concurrent communications via carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating over portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

(36) Signal waveforms transmitted over a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both) such that the more resource elements that a device receives and the higher the order of the modulation scheme, the higher the data rate may be for the device. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

(37) One or more numerologies for a carrier may be supported, where a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and

communications for the UE **115** may be restricted to one or more active BWPs.

(38) The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of T.sub.s=1/(Δf.sub.max.Math.N.sub.f) seconds, where Δf.sub.max may represent the maximum supported subcarrier spacing, and N.sub.f may represent the maximum supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

(39) Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems **100**, a slot may further be divided into multiple mini-slots containing one or more symbols. Excluding the cyclic prefix, each symbol period may contain one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

(40) A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).

(41) Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**.

(42) A network entity **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a network entity **105** (e.g., over a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell may also refer to a coverage area **110** or a portion of a coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the network entity **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with coverage areas **110**, among other examples.

(43) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered network entity **105** (e.g., a lower-powered base station **140**), as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A network entity **105** may support one or multiple cells and may also support communications over the one or more cells using one or multiple component carriers.
(44) In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.
(45) In some examples, a network entity **105** (e.g., a base station **140**, an RU **170**) may be movable and therefore provide communication coverage for a moving coverage area **110**. In some examples, different coverage areas **110** associated with different technologies may overlap, but the different coverage areas **110** may be supported by the same network entity **105**. In some other examples, the overlapping coverage areas **110** associated with different technologies may be supported by different network entities **105**. The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** provide coverage for various coverage areas **110** using the same or different radio access technologies.
(46) The wireless communications system **100** may support synchronous or asynchronous operation. For synchronous operation, network entities **105** (e.g., base stations **140**) may have similar frame timings, and transmissions from different network entities **105** may be approximately aligned in time. For asynchronous operation, network entities **105** may have different frame timings, and transmissions from different network entities **105** may, in some examples, not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.
(47) Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a network entity **105** (e.g., a base station **140**) without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that makes use of the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging. In an aspect, techniques disclosed herein may be applicable to MTC or IoT UEs. MTC or IoT UEs may include MTC/enhanced MTC (eMTC, also referred to as CAT-M, Cat M1) UEs, NB-IoT (also referred to as CAT NB1) UEs, as well as other types of UEs. eMTC and NB-IoT may refer to future technologies that may evolve from or may be based on these technologies. For example, eMTC may include FeMTC (further eMTC), eFeMTC (enhanced further eMTC), and mMTC (massive MTC), etc., and NB-IoT may include eNB-IoT (enhanced NB-IoT), and FeNB-IoT (further enhanced NB-IoT).
(48) Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via

transmission or reception, but not transmission and reception concurrently). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for the UEs **115** include entering a power saving deep sleep mode when not engaging in active communications, operating over a limited bandwidth (e.g., according to narrowband communications), or a combination of these techniques. For example, some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a defined portion or range (e.g., set of subcarriers or resource blocks (RBs)) within a carrier, within a guard-band of a carrier, or outside of a carrier.

(49) The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

(50) In some examples, a UE **115** may be able to communicate directly with other UEs **115** over a device-to-device (D2D) communication link **135** (e.g., in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (e.g., a base station **140**, an RU **170**), which may support aspects of such D2D communications being configured by or scheduled by the network entity **105**. In some examples, one or more UEs **115** in such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some examples, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in which each UE **115** transmits to each of the other UEs **115** in the group. In some examples, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs **115** without the involvement of a network entity **105**.

(51) In some systems, a D2D communication link **135** may be an example of a communication channel, such as a sidelink communication channel, between vehicles (e.g., UEs **115**). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may communicate with roadside infrastructure, such as roadside units, or with the network via one or more network nodes (e.g., network entities **105**, base stations **140**, RUs **170**) using vehicle-to-network (V2N) communications, or with both.

(52) The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (e.g., base stations **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be

connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

(53) The wireless communications system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. The UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. The transmission of UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than 100 kilometers) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

(54) The wireless communications system **100** may also operate in a super high frequency (SHF) region using frequency bands from 3 GHz to 30 GHz, also known as the centimeter band, or in an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, the wireless communications system **100** may support millimeter wave (mmW) communications between the UEs **115** and the network entities **105** (e.g., base stations **140**, RUs **170**), and EHF antennas of the respective devices may be smaller and more closely spaced than UHF antennas. In some examples, this may facilitate use of antenna arrays within a device. The propagation of EHF transmissions, however, may be subject to even greater atmospheric attenuation and shorter range than SHF or UHF transmissions. The techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.

(55) The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating in unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations in unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

(56) A network entity **105** (e.g., a base station **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located in diverse geographic locations. A network entity **105** may have an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may have one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

(57) The network entities **105** or the UEs **115** may use MIMO communications to exploit multipath signal propagation and increase the spectral efficiency by transmitting or receiving multiple signals

via different spatial layers. Such techniques may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry information associated with the same data stream (e.g., the same codeword) or different data streams (e.g., different codewords). Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO), where multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO), where multiple spatial layers are transmitted to multiple devices.
(58) Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).
(59) A network entity **105** or a UE **115** may use beam sweeping techniques as part of beamforming operations. For example, a network entity **105** (e.g., a base station **140**, an RU **170**) may use multiple antennas or antenna arrays (e.g., antenna panels) to conduct beamforming operations for directional communications with a UE **115**. Some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a network entity **105** multiple times along different directions. For example, the network entity **105** may transmit a signal according to different beamforming weight sets associated with different directions of transmission. Transmissions along different beam directions may be used to identify (e.g., by a transmitting device, such as a network entity **105**, or by a receiving device, such as a UE **115**) a beam direction for later transmission or reception by the network entity **105**.
(60) Some signals, such as data signals associated with a particular receiving device, may be transmitted by transmitting device (e.g., a transmitting network entity **105**, a transmitting UE **115**) along a single beam direction (e.g., a direction associated with the receiving device, such as a receiving network entity **105** or a receiving UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based on a signal that was transmitted along one or more beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the network entity **105** along different directions and may report to the network entity **105** an indication of the signal that the UE **115** received with a highest signal quality or an otherwise acceptable signal quality.
(61) In some examples, transmissions by a device (e.g., by a network entity **105** or a UE **115**) may be performed using multiple beam directions, and the device may use a combination of digital precoding or beamforming to generate a combined beam for transmission (e.g., from a network entity **105** to a UE **115**). The UE **115** may report feedback that indicates precoding weights for one or more beam directions, and the feedback may correspond to a configured set of beams across a system bandwidth or one or more sub-bands. The network entity **105** may transmit a reference signal (e.g., a cell-specific reference signal (CRS), a channel state information reference signal (CSI-RS)), which may be precoded or unprecoded. The UE **115** may provide feedback for beam

selection, which may be a precoding matrix indicator (PMI) or codebook-based feedback (e.g., a multi-panel type codebook, a linear combination type codebook, a port selection type codebook). Although these techniques are described with reference to signals transmitted along one or more directions by a network entity **105** (e.g., a base station **140**, an RU **170**), a UE **115** may employ similar techniques for transmitting signals multiple times along different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**) or for transmitting a signal along a single direction (e.g., for transmitting data to a receiving device).

(62) A receiving device (e.g., a UE **115**) may perform reception operations in accordance with multiple receive configurations (e.g., directional listening) when receiving various signals from a receiving device (e.g., a network entity **105**), such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may perform reception in accordance with multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets (e.g., different directional listening weight sets) applied to signals received at multiple antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at multiple antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive configurations or receive directions. In some examples, a receiving device may use a single receive configuration to receive along a single beam direction (e.g., when receiving a data signal). The single receive configuration may be aligned along a beam direction determined based on listening according to different receive configuration directions (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio (SNR), or otherwise acceptable signal quality based on listening according to multiple beam directions).

(63) The wireless communications system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or PDCP layer may be IP-based. An RLC layer may perform packet segmentation and reassembly to communicate over logical channels. A MAC layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use error detection techniques, error correction techniques, or both to support retransmissions at the MAC layer to improve link efficiency. In the control plane, the RRC protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a network entity **105** or a core network **130** supporting radio bearers for user plane data. At the PHY layer, transport channels may be mapped to physical channels.

(64) The UEs **115** and the network entities **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly over a communication link (e.g., a communication link **125**, a D2D communication link **135**). HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In some other examples, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

(65) For example, a reduced capability UE may receive an indication of multiple random access occasions that may be associated with one or more initial UL BWPs that may be used for performing a random access procedure. In some examples, each initial UL BWP may include a corresponding random access occasion, while in other examples some of the multiple random access occasions may be located in frequency resources outside of the initial UL BWP. The UE may select a random access occasion of the multiple random access occasions and may transmit a

random access preamble (e.g., Msg1 of a random access channel (RACH) procedure) within the random access occasion.

(66) FIGS. **2**A and **2**B illustrate examples of a bandwidth part configuration **200** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. In the descriptions of the bandwidth part configuration **200**, the BWPs **205** may be initial UL BWPs used in associated with random access procedures for eRedCap devices and the ROs **210** may be random access occasions used for such random access procedures.

(67) Some wireless communications network may employ the use of reduced capability (RedCap) or enhanced reduced capability (eRedCap) devices, which may operate with reduced capabilities such as reduced peak data throughput, latency, reliability, bandwidth, or other characteristics or capabilities. For example, some eRedCap devices may operate within a reduced bandwidth (e.g., 5 MHz). Such characteristics may limit a number of random access occasions that may be multiplexed (e.g., via FDM) and may lead to collisions during a random access procedure. For example, in some approaches, an initial UL BWP for eRedCap devices may contain one or two random access occasions with a 30 or 15 kHz subcarrier spacing (SCS) and a short sequence. Such approaches may lead to a limited number of allowed simultaneous random access occasions in the time domain (e.g., 64 random access occasions), which may lead to collisions.

(68) In some approaches, eRedCap devices may use longer durations (for CORESETs or PDSCH messages) than durations used by other devices to attain similar coverage levels. Such operations or characteristics may also limit a number of eRedCap UEs whose paging occasions may be accommodated within an initial UL BWP for eRedCap devices. Some approaches may attempt to resolve such issues by employing TDM to offer additional random access occasions to eRedCap devices (which may result in increased overhead) or may use lengthened physical random access channel (PRACH) formats (which may result in an SCS that may be different that an SCSC used for data communications).

(69) Therefore, to reduce or eliminate such limitations, a network entity may configure multiple initial BWPs (e.g., initial UL BWPs, initial DL BWPs, or both) for eRedCap devices and may further map multiple random access occasions (e.g., using FDM) to an initial UL BWP. Such approaches may offer increased capacity for eRedCap devices while reducing collisions during random access procedures.

(70) As depicted in FIG. **2**A, a network may configure multiple initial UL BWPs, such as BWP **205**-*a*, **205**-*b*, **205**-*c*, and **205**-*d* for eRedCap devices to multiplex random access loads. In some examples, each BWP **205** may contain a random access occasion (e.g., a RACH occasion (RO)), such as RO **210**-*a*, RO **210**-*b*, RO **210**-*c*, and RO **210**-*d*. For example, an RO **210** may be located within frequency resources that are located within a corresponding BWP **205**. A UE may select one of the ROs **210** to use for performing a random access procedure (e.g., a RACH procedure), which may determine an effective initial UL BWP, an effective initial DL BWP, or both for the eRedCap UE to use for a random access procedure. For example, a eRedCap UE may select RO **210**-*b* for communicating a random access preamble. By selecting RO **210**-*b*, the UE, a network entity, or both, may determine that subsequent messaging (e.g., for a random access procedure, such as Msg2, Msg3, Msg4, or other RACH procedure signaling, other subsequent signaling, or any combination thereof) is to be performed within the BWP **205**-*b* that corresponds to or contains the selected RO **210**-*b*. In this way, capacity for serving multiple eRedCap devices may be increased and interference (e.g., collisions during a random access procedure) may be reduced or eliminated.

(71) In the course of a random access procedure, transmission or reception of a random access preamble (e.g., Msg1 of a RACH procedure) may fail. In such cases, the eRedCap UE may employ one or more options for power control for retransmitting a failed random access preamble message. For example, the eRedCap UE may employ one or more power control parameters (e.g., a power ramping counter, such as a preamble transmission power ramping counter, a maximum power

ramping parameter or value, such as a preamble transmission power maximum parameter or value, or any combination thereof) for retransmitting a failed random access preamble message.

(72) In some examples, if a random access preamble transmission fails, a UE may select an RO **210** for retransmission of a failed random access preamble message. In such examples, the UE may select an RO from the same BWP **205** that was used during a prior transmission (e.g., during the failed transmission of the random access preamble) and the UE may increase or increment a preamble transmission power ramping counter based on such a selection. Additionally, or alternatively, the UE may select an RO from a different BWP **205** than the BWP **205** that was used during a prior transmission (e.g., during the failed transmission of the random access preamble) and the UE may refrain from increasing a preamble transmission power ramping counter based upon such a selection. Additionally, or alternatively, a UE may increment or increase a preamble transmission power ramping counter regardless of whether a selected RO **210** falls within a same BWP **205** that was used during a prior transmission (e.g., during the failed transmission of the random access preamble).

(73) In some examples, the UE may be configured with one or more global power control parameters for preamble transmissions, one or more local parameters for preamble transmissions, or any combination thereof. For example, a UE may be configured with a global preamble transmission counter and preamble_transmission_max. The UE may select either a same or different BWP **205** used during a prior transmission (e.g., during the failed transmission of the random access preamble) for retransmission of a random access preamble message during a random access procedure. The UE may increment the global preamble_transmission_counter (e.g., using approaches described herein regarding selecting a same BWP **205** or a different BWP **205** than one used previously and incrementing or not incrementing based on such a selection) until the transmission of the preamble is successful or preamble_transmission_counter has a value that is equal to a value of preamble_transmission_max. Additionally, or alternatively, a UE may be configured with one or more local parameters, such a one or more instances of a preamble_transmission_counter, preamble_transmission_max, or any combination thereof. In some examples, such local parameters may be configured, incremented, adjusted, or otherwise utilized on a per-BWP basis. For example, a UE may have a preamble_transmission_counter, preamble_transmission_max, or both for one or more available BWPs (e.g., for one or more of BWP **205**-*a*, BWP **205**-*b*, BWP **205**-*c*, and BWP **205**-*d*). In some examples, a UE may select the same BWP **205** used during a prior transmission (e.g., during the failed transmission of the random access preamble) and may increment the preamble_transmission_counter (e.g., using approaches described herein regarding selecting a same BWP **205** or a different BWP **205** than one used previously and incrementing or not incrementing based on such a selection) for each retransmission until the transmission of the preamble is successful or preamble_transmission_counter for a BWP **205** has a value that is equal to a value of preamble_transmission_max for the same BWP **205**.

(74) As depicted in FIG. **2**B, a network entity may configure a single BWP **205**-*a* and multiple ROs **210** that may be associated with the BWP **205**-*a* (e.g., to allow a greater number of ROs to be made available to eRedCap devices while maintaining a single initial UL BWP). In some examples, one or more of the ROs (e.g., RO **210**-*a*) may be located in frequency resources that fall within the BWP **205**-*a*, whereas other ROs (e.g., RO **210**-*b*, RO **210**-*c*, and RO **210**-*d*) may be located in frequency resources that fall outside of the BWP **205**-*a* but may still be associated with the BWP **205**-*a*. In some such examples, a bandwidth of an RO **210** may be less than or equal to a bandwidth value used for an RO **210** that falls within the BWP **205**-*a* (e.g., 5 MHz), but the overall bandwidth of the multiple ROs **210** may be wider or greater than a bandwidth associated with the BWP **205**-*a*. A UE may select (e.g., regardless of whether a selected RO falls within the BWP **205** or not) any of the ROs **210** for communications such as transmission of a random access preamble or Msg1 of a RACH preamble. In some examples, subsequent messaging (e.g., Msg2, Msg3, Msg4, other messages, or any combination thereof) may be communicated within the BWP **205**-*a*, whether the

selected RO **210** falls within the BWP **205**-*a* or not.

(75) In some examples, parameters associated with the random access procedure or RACH procedure may be modified or new parameters may be used. For example, an offset parameter (e.g., msg1FrequencyStart associated with a RACH procedure) may be modified in implementations (e.g., the implementation depicted in FIG. **2**B). Such an offset parameter may indicate an offset of a lowest random access transmission occasion (e.g., an RO **210**) in the frequency domain with respect to a reference point or range (e.g., a physical resource block (PRB) such as PRB 0, that may be associated with the BWP **205**-*a*). In some examples, such a parameter may be modified to remove a restriction that a random access resource is contained entirely within a bandwidth of a BWP, such as BWP **205**-*a*. For example, an offset parameter (e.g., msg1FrequencyStart) may indicate that a lowest random access transmission occasion (e.g., an RO **210**) may be partially or completely located outside a bandwidth of the BWP **205**-*a*.

(76) In some examples, a UE that selects an RO **210** that falls outside of the BWP **205**-*a* (e.g., RO **210**-*c*) may effectively be performing a BWP switch in a time domain (e.g., via TDD) between transmission of a random access preamble (e.g., RACH Msg1) and other random access procedure messaging (e.g., RACH procedure messaging such as Msg2, Msg3, Msg4, etc.). In some such situations, a network entity may configure or designate a period of time for switching between bandwidth parts at one or more points in a random access process (e.g., between a RACH Msg1 and Msg2, between a RACH Msg2 and Msg3, or at other places in a random access procedure). For example, a first group of UEs (such as UEs with a single phase-locked loop (PLL)) may utilize BWP switching time between transmitting a random access preamble (e.g., RACH Msg1) and receiving a subsequent random access message (e.g., RACH Msg2). Some approaches may include a gap that is greater than, less than, or equal to a designated time gap (e.g., a gap of one symbol). In another example, a second group of UEs may not utilize a time gap between transmitting a random access preamble (e.g., RACH Msg1) and receiving a subsequent random access message (e.g., RACH Msg2), but may utilize a time gap between transmission of a random access preamble (e.g., RACH Msg1) and transmitting a subsequent random access message (e.g., RACH Msg3). Additionally, or alternatively, the second group of UEs may utilize a time gap between receiving a random access message (e.g., RACH Msg2) and transmitting a subsequent random access message (e.g., RACH Msg3). Some approaches may include a gap that is greater than, less than, or equal a designated time gap or threshold. Though examples of gaps are discussed, the subject matter described herein contemplates such time gaps at any (or multiple) points (e.g., between any pairs of random access messages, whether they are sequential in time or not) during a random access procedure.

(77) In some examples, the UE may transmit a random access preamble **215** during one of the ROs **210** (which may be located within the BWP **205**-*a* or may be located outside of the BWP **205**-*a*, which itself may be a shared initial UL BWP for eRedCap UEs), and may communicate a random access payload **220** within the BWP **205**-*a*. In some examples, and as discussed herein, a time gap may be configured or designated between communication of the preamble **215** and the payload **220** to allow the UE to perform a BWP switch or one or more other RF retuning procedures. In some examples, if multiple preambles on the same RO **210** are mapped to the same PUSCH occasion, the same gap between (e.g., between a msgA preamble and a msgA payload) may be configured to promote consistency of timing advances. Additionally, or alternatively, if multiple initial UL BWPs are supported, repetition and inter-slot frequency hopping (e.g., of msgA preambles, payloads, or both) may be configured.

(78) FIG. **3** illustrates an example of a bandwidth part configuration **300** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The bandwidth part configuration **300** may include or relate to a non-RedCap initial UL BWP **310**, a non-RedCap initial DL BWP **315**, an eRedCap initial DL BWP **320**, one or more eRedCap initial UL BWPs **325** (including, as examples, eRedCap initial UL BWP **325**-*a* and

eRedCap initial UL BWP **325**-*b*), one or more eRedCap initial DL BWPs **330** (including, as examples, eRedCap initial DL BWP **330**-*a* and eRedCap initial DL BWP **330**-*b*), or any combination thereof.

(79) In some examples, a network entity may configure a non-RedCap initial UL BWP **310** and a non-RedCap initial DL BWP **315** for non-RedCap devices. The non-RedCap initial DL BWP **315** may include a synchronization signal block (SSB), such as a cell-defining SSB (CD-SSB), a control resource set (CORESET) such as CORESET0, or any combination thereof. Such parameters, values, or information may be used at least by non-RedCap UEs.

(80) The network entity may configure an eRedCap initial DL BWP **320**, which may include a modified SSB (e.g., a modified CD-SSB), a modified CORESET (e.g., a modified CORESET0), a system information block (SIB), paging information, or any combination thereof, that may be used by eRedCap UEs for random access procedures (e.g., a RACH procedure). For example, an SSB (e.g., the modified CD-SSB) may include a SIB that may include information about one or more initial UL BWPs, one or more initial DL BWPs, one or more ROs, or any combination thereof that a UE may utilize for a random access procedure. For example, the SSB may include information indicating one or more eRedCap initial UL BWPs **325** and one or more eRedCap initial DL BWPs **330**. An eRedCap initial UL BWPs **325** may be configured to avoid fragmentation of uplink resources of non-eRedCap UEs. Further, an eRedCap initial DL BWP **330** may be configured to align a random access common search space (CSS) with an initial UL BWP (e.g., an eRedCap initial UL BWP **325**).

(81) In some examples, an initial DL BWP (e.g., an eRedCap initial DL BWP **330**) may be tied to or associated with a paging location of a UE (e.g., based on the UE selecting an RO associated with an eRedCap initial UL BWP **325** that may be associated with an eRedCap initial DL BWP **330**). In some examples, a UE may monitor paging during a random access procedure. In such cases, a UE may select the same eRedCap initial UL BWP **325** that is associated with the eRedCap initial DL BWP **330** that contains the UE's paging location. For example, if a paging location for the UE is located in eRedCap initial DL BWP **330**-*a*, the UE may select the eRedCap initial UL BWP **325**-*a* for a random access procedure (e.g., since the eRedCap initial DL BWP **330**-*a* is associated with the initial UL BWP **325**-*a*). However, if a UE is not monitoring paging during a random access procedure, the UE may select any eRedCap initial UL BWP **325** for the random access procedure (e.g., for transmitting or retransmitting a random access preamble).

(82) Additionally, or alternatively, an eRedCap initial DL BWP **330** may be configured with a modified non cell-defining SSB (NCD-SSB), paging search spaces, or both, that a eRedCap UE may utilize (e.g., during or associated with a random access procedure). Additionally, or alternatively, a mapping between an identifier of a UE and an identifier of an initial BWP may be created or utilized for finding a paging occasion.

(83) FIGS. **4**A, **4**B, and **4**C illustrate examples of a bandwidth part configuration **400** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.

(84) Some approaches for random access procedures are limited in their allocation of random access occasions. For example, in some approaches, random access occasions are allocated in contiguous blocks and may extend in a single direction from a reference point (e.g., a PRB 0 of a BWP). In such approaches, an offset parameter (e.g., msg1-FrequencyStart) may indicate an offset between a random access occasion and the reference point, and other random access occasions may be indicated through another parameter (e.g., msg1-FDM). However, offsets of such random access occasions for a BWP may not be referred to relative to an absolute frequency position of a reference resource block (e.g., a parameter such as absoluteFrequencyPointA that may indicate an absolute frequency position of a reference resource block such as a common RB 0). However, such approaches may be improved.

(85) As depicted in FIG. **4**A, a binary direction parameter may be introduced. The direction

parameter may indicate a direction **415** (e.g., in a frequency domain) that may indicate in which direction from a reference point (e.g., an RO indicated by an offset parameter which may indicate the offset **420**) a contiguous block of ROs may be allocated. The direction parameter may be used in combination with other parameters (e.g., a multiplexing parameter, (e.g., msg1-FDM), an offset parameter (e.g., msg1-FrequencyStart, one or more other parameters, or any combination thereof) to determine or select multiple ROs that may be available to an eRedCap UE for use in connection with a random access procedure. The offset parameter may indicate an offset **420** from a reference point (e.g., PRB 0) of an initial UL BWP (e.g., BWP **405**-*a*) to an RO **410** (e.g., RO **410**-*a*, as depicted in FIG. **4**A). The direction parameter may, based on the offset **420**, then indicate the direction **415** (e.g., up or down in frequency) in which other ROs (e.g., ROs **410**-*b*, **410**-*c*, and **410**-*d*) may be allocated. These other ROs may be indicated or defined by the multiplexing parameter (e.g., msg1-FDM).
(86) Such approaches may provide for additional flexibility, in that the ROs **410** associated with the BWP **405**-*a* may extend upwards or downwards in the frequency domain. Such ROs **410** may be allocated in a contiguous fashion, and in some examples, at least one RO **410** may be located in a positive offset (e.g., increasing in the frequency domain) from a reference point (e.g., PRB 0) of the BWP **405**-*a*.
(87) As depicted in FIG. **4**B, multiple offset parameters may be provided that may indicate offsets **420** (e.g., including offset **420**-*a*, offset **420**-*b*, offset **420**-*c*, and offset **420**-*d*) for each of the ROs **410** on an individual or per-RO basis. For example, a network entity may configure multiple offset parameters (e.g., msg1-FrequencyStart-RO parameters) for some or all of the ROs **410** associated with the BWP **405**-*a*. The offsets **420** indicates by the corresponding offset parameters may be positive or negative for each associated RO **410**. In this way, the network entity may configure multiple ROs in different locations in the frequency domain, and may do so irrespective of a location in the frequency domain of the BWP **405**-*a*.
(88) As depicted in FIG. **4**C, a new reference point (e.g., a Point A, as may be indicated in a parameter such as absoluteFrequencyPointA that may indicate an absolute frequency position of a reference resource block) for a first RO of the ROs **410** may be selected, indicated, or determined. Such a reference point may be used instead of another reference point (such as PRB 0 of the BWP **405**-*a*), and an offset parameter may indicate an offset **420** from the reference point. In some examples, additional ROs, such as ROs **410**-*b*, **410**-*c*, and **410**-*d*, may be determined or allocated based on the position of a first RO, such as RO **410**-*a* (e.g., as described herein, such as in connection with a multiplexing parameter such as msg1-FDM). In such approaches, any RO **410** may be located in a positive or negative offset (relative to a point of or associated with the BWP **405**-*a*, such as PRB 0) since a different reference point is used. Further, signaling overhead may be decreased as compared to other approaches.
(89) FIG. **5** illustrates an example of a process flow **500** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure.
(90) The process flow **500** may implement various aspects of the present disclosure described herein. The network entity **105**-*a* and the UE **115**-*a* described in the process flow **500** may be examples of similarly-named elements described herein. In the following description of the process flow **500**, the operations between the various entities or elements may be performed in different orders or at different times. Some operations may also be left out of the process flow **500**, or other operations may be added. Although the various entities or elements are shown performing the operations of the process flow **500**, some aspects of some operations may also be performed by other entities or elements of the process flow **500** or by entities or elements that are not depicted in the process flow, or any combination thereof.
(91) At **520**, the UE **115**-*a* may receive a message that may indicate a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also optionally indicating a plurality of random access occasions, at least one of the plurality of random access

occasions being in frequency resources that are outside of the first initial uplink bandwidth part. In some examples, receiving the message may include receiving an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the plurality of random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts. In some examples, at most one of the plurality of the random access occasions may be within the first initial uplink bandwidth part, a remainder of the plurality of the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.
(92) At **525**, the UE **115**-*a* may receive information that may be indicative of at least one of an offset parameter or a binary direction parameter and the binary direction parameter may indicate a direction in which the offset parameter may be applied with respect to a baseline resource block. In some examples, the UE **115**-*a* may receive indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis. The UE **115**-*a* may determine one or more of the frequency resources that may be outside of the first initial uplink bandwidth part based at least in part on the information. In some examples, determining the one or more of the frequency resources that may be outside of the first initial uplink bandwidth part may include determining, based at least in part on the offset parameter, the binary direction parameter, the baseline resource block, and a first frequency resource associated with a first random access occasion of the plurality of random access occasions and determining additional frequency resources associated with other random access occasions of the plurality of random access occasions, the additional frequency resources determined in relation to the first frequency resource.
(93) At **530**, the UE **115**-*a* may select a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity. In some examples, the UE **115**-*a* may monitor an initial downlink bandwidth part for paging occasions during the random access procedure. The UE **115**-*a* may identify, from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, a corresponding initial uplink bandwidth part that corresponds to the initial downlink bandwidth part. The UE **115**-*a* may further select the random access occasion that may be within the corresponding initial uplink bandwidth part based at least in part on the corresponding initial uplink bandwidth part corresponding to the initial downlink bandwidth part.
(94) At **535**, the UE **115**-*a* may transmit, as part of a random access procedure, the random access preamble within the random access occasion. In some examples, the UE **115**-*a* may determine that transmission of the random access preamble failed and that the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts. The UE **115**-*a* may increment a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble and on whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble. Additionally, or alternatively, the UE **115**-*a* may increment a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble, regardless of whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble.
(95) The UE **115**-*a* may retransmit the random access preamble in accordance with the preamble power ramping parameter. Additionally, or alternatively, the UE **115**-*a* may retransmit the random access preamble via the first attempted initial uplink bandwidth part and the retransmitting may be in accordance with a maximum number of retransmissions parameter that applies to retransmissions per initial uplink bandwidth part.
(96) In some examples, the UE **115**-*a* may determine that transmission of the random access preamble failed and may further retransmit the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts and the retransmitting may be in accordance with a maximum number of retransmissions parameter that applies to

retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.
(97) At **540**, the UE **115**-*a* may communicate additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble may be transmitted.
(98) Additionally, or alternatively, the UE **115**-*a* may communicate additional one or more random access messages that may be associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that may be within a selected initial uplink bandwidth part that corresponds to the random access occasion. In some examples, communicating the additional one or more random access messages may include receiving a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap. In some examples, communicating the additional one or more random access messages may include transmitting transmit a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap. In some examples, communicating the additional one or more random access messages may include transmitting a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.
(99) FIG. **6** shows a block diagram **600** of a device **605** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **605** may be an example of aspects of a UE **115** as described herein. The device **605** may include a receiver **610**, a transmitter **615**, and a communications manager **620**. The device **605** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(100) The receiver **610** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to bandwidth part configuration for reduced capability devices). Information may be passed on to other components of the device **605**. The receiver **610** may utilize a single antenna or a set of multiple antennas.
(101) The transmitter **615** may provide a means for transmitting signals generated by other components of the device **605**. For example, the transmitter **615** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to bandwidth part configuration for reduced capability devices). In some examples, the transmitter **615** may be co-located with a receiver **610** in a transceiver module. The transmitter **615** may utilize a single antenna or a set of multiple antennas.
(102) The communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.
(103) In some examples, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other

programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

(104) Additionally, or alternatively, in some examples, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in code (e.g., as communications management software) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, a GPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

(105) In some examples, the communications manager **620** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **610**, the transmitter **615**, or both. For example, the communications manager **620** may receive information from the receiver **610**, send information to the transmitter **615**, or be integrated in combination with the receiver **610**, the transmitter **615**, or both to obtain information, output information, or perform various other operations as described herein.

(106) Additionally, or alternatively, the communications manager **620** may support wireless communications at a UE in accordance with examples as disclosed herein. For example, the communications manager **620** may be configured as or otherwise support a means for receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The communications manager **620** may be configured as or otherwise support a means for selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The communications manager **620** may be configured as or otherwise support a means for transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

(107) By including or configuring the communications manager **620** in accordance with examples as described herein, the device **605** (e.g., a processor controlling or otherwise coupled with the receiver **610**, the transmitter **615**, the communications manager **620**, or a combination thereof) may support techniques for reduced processing, reduced power consumption, more efficient utilization of communication resources, or any combination thereof.

(108) FIG. **7** shows a block diagram **700** of a device **705** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **705** may be an example of aspects of a device **605** or a UE **115** as described herein. The device **705** may include a receiver **710**, a transmitter **715**, and a communications manager **720**. The device **705** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(109) The receiver **710** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to bandwidth part configuration for reduced capability devices). Information may be passed on to other components of the device **705**. The receiver **710** may utilize a single antenna or a set of multiple antennas.

(110) The transmitter **715** may provide a means for transmitting signals generated by other components of the device **705**. For example, the transmitter **715** may transmit information such as

packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to bandwidth part configuration for reduced capability devices). In some examples, the transmitter **715** may be co-located with a receiver **710** in a transceiver module. The transmitter **715** may utilize a single antenna or a set of multiple antennas.
(111) The device **705**, or various components thereof, may be an example of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **720** may include a BWP message reception component **725**, a RO selection component **730**, a random access preamble transmission component **735**, or any combination thereof. The communications manager **720** may be an example of aspects of a communications manager **620** as described herein. In some examples, the communications manager **720**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **710**, the transmitter **715**, or both. For example, the communications manager **720** may receive information from the receiver **710**, send information to the transmitter **715**, or be integrated in combination with the receiver **710**, the transmitter **715**, or both to obtain information, output information, or perform various other operations as described herein.
(112) The communications manager **720** may support wireless communications at a UE in accordance with examples as disclosed herein. The BWP message reception component **725** may be configured as or otherwise support a means for receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The RO selection component **730** may be configured as or otherwise support a means for selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The random access preamble transmission component **735** may be configured as or otherwise support a means for transmitting, as part of a random access procedure, the random access preamble within the random access occasion.
(113) FIG. **8** shows a block diagram **800** of a communications manager **820** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The communications manager **820** may be an example of aspects of a communications manager **620**, a communications manager **720**, or both, as described herein. The communications manager **820**, or various components thereof, may be an example of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **820** may include a BWP message reception component **825**, a RO selection component **830**, a random access preamble transmission component **835**, a random access procedure communication component **840**, a power parameter component **845**, a bandwidth part component **850**, a frequency resource parameter component **855**, a frequency resource selection component **860**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).
(114) Additionally, or alternatively, the communications manager **820** may support wireless communications at a UE in accordance with examples as disclosed herein. The BWP message reception component **825** may be configured as or otherwise support a means for receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The RO selection component **830** may be configured as or otherwise support a means for selecting a random access occasion of the set of

multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The random access preamble transmission component **835** may be configured as or otherwise support a means for transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

(115) In some examples, to support receiving the message, the BWP message reception component **825** may be configured as or otherwise support a means for receiving an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the set of multiple random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.

(116) In some examples, the random access procedure communication component **840** may be configured as or otherwise support a means for communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble is transmitted.

(117) In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts. In some examples, the power parameter component **845** may be configured as or otherwise support a means for incrementing a preamble power ramping parameter based on failure of the transmission of the random access preamble and on whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble. In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for retransmitting the random access preamble in accordance with the preamble power ramping parameter.

(118) In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts. In some examples, the power parameter component **845** may be configured as or otherwise support a means for incrementing a preamble power ramping parameter based on failure of the transmission of the random access preamble, regardless of whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble. In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for retransmitting the random access preamble in accordance with the preamble power ramping parameter.

(119) In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for determining that transmission of the random access preamble failed. In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for retransmitting the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, where the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.

(120) In some examples, the random access preamble transmission component **835** may be configured as or otherwise support a means for determining that transmission of the random access preamble failed, where the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts. In some examples, the random access preamble transmission component

835 may be configured as or otherwise support a means for retransmitting the random access preamble via the first attempted initial uplink bandwidth part, where the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions per initial uplink bandwidth part.
(121) In some examples, to support selecting the random access occasion, the bandwidth part component **850** may be configured as or otherwise support a means for monitoring an initial downlink bandwidth part for paging occasions during the random access procedure. In some examples, to support selecting the random access occasion, the bandwidth part component **850** may be configured as or otherwise support a means for identifying, from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, a corresponding initial uplink bandwidth part that corresponds to the initial downlink bandwidth part. In some examples, to support selecting the random access occasion, the bandwidth part component **850** may be configured as or otherwise support a means for selecting the random access occasion that is within the corresponding initial uplink bandwidth part based on the corresponding initial uplink bandwidth part corresponding to the initial downlink bandwidth part.
(122) In some examples, at most one of the set of multiple the random access occasions is within the first initial uplink bandwidth part, a remainder of the set of multiple the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.
(123) In some examples, the frequency resource parameter component **855** may be configured as or otherwise support a means for receiving information that is indicative of at least one of an offset parameter or a binary direction parameter, where the binary direction parameter indicates a direction in which the offset parameter is to be applied with respect to a baseline resource block. In some examples, the frequency resource selection component **860** may be configured as or otherwise support a means for determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part based on the information.
(124) In some examples, to support receiving the information that is indicative of at least one of the offset parameter or the binary direction parameter, the frequency resource selection component **860** may be configured as or otherwise support a means for receiving indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis.
(125) In some examples, to support determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part, the frequency resource parameter component **855** may be configured as or otherwise support a means for determining, based on the offset parameter, the binary direction parameter, the baseline resource block, and a first frequency resource associated with a first random access occasion of the set of multiple random access occasions. In some examples, to support determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part, the frequency resource selection component **860** may be configured as or otherwise support a means for determining additional frequency resources associated with other random access occasions of the set of multiple random access occasions, the additional frequency resources determined in relation to the first frequency resource.
(126) In some examples, the random access procedure communication component **840** may be configured as or otherwise support a means for communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.
(127) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication component **840** may be configured as or otherwise support a means for receiving a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

(128) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication component **840** may be configured as or otherwise support a means for transmitting a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(129) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication component **840** may be configured as or otherwise support a means for transmitting a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.
(130) In some examples, the message is received through a system information block, a synchronization signal block, downlink control information, a medium access control control element, radio resource control signaling, a handover command, or any combination thereof.
(131) FIG. **9** shows a diagram of a system **900** including a device **905** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **905** may be an example of or include the components of a device **605**, a device **705**, or a UE **115** as described herein. The device **905** may communicate (e.g., wirelessly) with one or more network entities **105**, one or more UEs **115**, or any combination thereof. The device **905** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **920**, an input/output (I/O) controller **910**, a transceiver **915**, an antenna **925**, a memory **930**, code **935**, and a processor **940**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **945**).
(132) The I/O controller **910** may manage input and output signals for the device **905**. The I/O controller **910** may also manage peripherals not integrated into the device **905**. In some cases, the I/O controller **910** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **910** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally or alternatively, the I/O controller **910** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **910** may be implemented as part of a processor, such as the processor **940**. In some cases, a user may interact with the device **905** via the I/O controller **910** or via hardware components controlled by the I/O controller **910**.
(133) In some cases, the device **905** may include a single antenna **925**. However, in some other cases, the device **905** may have more than one antenna **925**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **915** may communicate bi-directionally, via the one or more antennas **925**, wired, or wireless links as described herein. For example, the transceiver **915** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **915** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **925** for transmission, and to demodulate packets received from the one or more antennas **925**. The transceiver **915**, or the transceiver **915** and one or more antennas **925**, may be an example of a transmitter **615**, a transmitter **715**, a receiver **610**, a receiver **710**, or any combination thereof or component thereof, as described herein.
(134) The memory **930** may include random access memory (RAM) and read-only memory (ROM). The memory **930** may store computer-readable, computer-executable code **935** including instructions that, when executed by the processor **940**, cause the device **905** to perform various functions described herein. The code **935** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **935** may not

be directly executable by the processor **940** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **930** may contain, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(135) The processor **940** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a GPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **940** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **940**. The processor **940** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **930**) to cause the device **905** to perform various functions (e.g., functions or tasks supporting bandwidth part configuration for reduced capability devices). For example, the device **905** or a component of the device **905** may include a processor **940** and memory **930** coupled with or to the processor **940**, the processor **940** and memory **930** configured to perform various functions described herein.

(136) Additionally, or alternatively, the communications manager **920** may support wireless communications at a UE in accordance with examples as disclosed herein. For example, the communications manager **920** may be configured as or otherwise support a means for receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The communications manager **920** may be configured as or otherwise support a means for selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The communications manager **920** may be configured as or otherwise support a means for transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

(137) By including or configuring the communications manager **920** in accordance with examples as described herein, the device **905** may support techniques for improved communication reliability, reduced latency, improved user experience related to reduced processing, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, improved utilization of processing capability, or any combination thereof.

(138) In some examples, the communications manager **920** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **915**, the one or more antennas **925**, or any combination thereof. Although the communications manager **920** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **920** may be supported by or performed by the processor **940**, the memory **930**, the code **935**, or any combination thereof. For example, the code **935** may include instructions executable by the processor **940** to cause the device **905** to perform various aspects of bandwidth part configuration for reduced capability devices as described herein, or the processor **940** and the memory **930** may be otherwise configured to perform or support such operations.

(139) FIG. **10** shows a block diagram **1000** of a device **1005** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **1005** may be an example of aspects of a network entity **105** as described herein. The device **1005** may include a receiver **1010**, a transmitter **1015**, and a communications manager **1020**. The device **1005** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(140) The receiver **1010** may provide a means for obtaining (e.g., receiving, determining,

identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **1005**. In some examples, the receiver **1010** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **1010** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

(141) The transmitter **1015** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **1005**. For example, the transmitter **1015** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **1015** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **1015** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **1015** and the receiver **1010** may be co-located in a transceiver, which may include or be coupled with a modem.

(142) The communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.

(143) In some examples, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a DSP, a CPU, a GPU, an ASIC, an FPGA or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

(144) Additionally, or alternatively, in some examples, the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be implemented in code (e.g., as communications management software) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **1020**, the receiver **1010**, the transmitter **1015**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, a GPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

(145) In some examples, the communications manager **1020** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1010**, the transmitter **1015**, or both. For example, the communications manager **1020** may receive information from the receiver **1010**, send information to the transmitter **1015**, or be integrated in combination with the receiver **1010**, the transmitter **1015**, or both to obtain information, output information, or perform various other operations as described herein.

(146) Additionally, or alternatively, the communications manager **1020** may support wireless communications at a network entity in accordance with examples as disclosed herein. For example, the communications manager **1020** may be configured as or otherwise support a means for transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The communications manager **1020** may be configured as or otherwise support a means for receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.

(147) By including or configuring the communications manager **1020** in accordance with examples as described herein, the device **1005** (e.g., a processor controlling or otherwise coupled with the receiver **1010**, the transmitter **1015**, the communications manager **1020**, or a combination thereof) may support techniques for reduced processing, reduced power consumption, more efficient utilization of communication resources, or any combination thereof.

(148) FIG. **11** shows a block diagram **1100** of a device **1105** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **1105** may be an example of aspects of a device **1005** or a network entity **105** as described herein. The device **1105** may include a receiver **1110**, a transmitter **1115**, and a communications manager **1120**. The device **1105** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(149) The receiver **1110** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **1105**. In some examples, the receiver **1110** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **1110** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

(150) The transmitter **1115** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **1105**. For example, the transmitter **1115** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **1115** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **1115** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **1115** and the receiver **1110** may be co-located in a transceiver, which may include or be coupled with a modem.

(151) The device **1105**, or various components thereof, may be an example of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **1120** may include a BWP message transmission element **1125** a random access preamble reception element **1130**, or any combination thereof. The communications manager **1120** may be an example of aspects of a communications manager **1020** as described herein. In some examples, the communications manager **1120**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1110**, the transmitter **1115**, or both. For example, the communications manager **1120** may receive

information from the receiver **1110**, send information to the transmitter **1115**, or be integrated in combination with the receiver **1110**, the transmitter **1115**, or both to obtain information, output information, or perform various other operations as described herein.
(152) The communications manager **1120** may support wireless communications at a network entity in accordance with examples as disclosed herein. The BWP message transmission element **1125** may be configured as or otherwise support a means for transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The random access preamble reception element **1130** may be configured as or otherwise support a means for receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.
(153) FIG. **12** shows a block diagram **1200** of a communications manager **1220** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The communications manager **1220** may be an example of aspects of a communications manager **1020**, a communications manager **1120**, or both, as described herein. The communications manager **1220**, or various components thereof, may be an example of means for performing various aspects of bandwidth part configuration for reduced capability devices as described herein. For example, the communications manager **1220** may include a BWP message transmission element **1225**, a random access preamble reception element **1230**, a random access procedure communication element **1235**, a frequency resource parameter element **1240**, a frequency resource element **1245**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses) which may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity **105**, between devices, components, or virtualized components associated with a network entity **105**), or any combination thereof.
(154) Additionally, or alternatively, the communications manager **1220** may support wireless communications at a network entity in accordance with examples as disclosed herein. The BWP message transmission element **1225** may be configured as or otherwise support a means for transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The random access preamble reception element **1230** may be configured as or otherwise support a means for receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.
(155) In some examples, to support receiving the message, the BWP message transmission element **1225** may be configured as or otherwise support a means for transmitting an indication of one or more additional initial uplink bandwidth parts for use by a UE in the communications with the network entity, each of the set of multiple random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.
(156) In some examples, the random access procedure communication element **1235** may be configured as or otherwise support a means for communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble is transmitted.
(157) In some examples, the random access preamble reception element **1230** may be configured as

or otherwise support a means for determining that transmission of the random access preamble failed. In some examples, the random access preamble reception element **1230** may be configured as or otherwise support a means for receiving a retransmission of the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, where the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.

(158) In some examples, at most one of the set of multiple the random access occasions is within the first initial uplink bandwidth part, a remainder of the set of multiple the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.

(159) In some examples, the frequency resource parameter element **1240** may be configured as or otherwise support a means for transmitting information that is indicative of at least one of an offset parameter or a binary direction parameter, where the binary direction parameter indicates a direction in which the offset parameter is to be applied with respect to a baseline resource block. In some examples, the frequency resource element **1245** may be configured as or otherwise support a means for determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part based on the information.

(160) In some examples, to support transmitting the information that is indicative of at least one of the offset parameter or the binary direction parameter, the frequency resource parameter element **1240** may be configured as or otherwise support a means for transmitting indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis.

(161) In some examples, the random access procedure communication element **1235** may be configured as or otherwise support a means for communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.

(162) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication element **1235** may be configured as or otherwise support a means for transmitting a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

(163) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication element **1235** may be configured as or otherwise support a means for receiving a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

(164) In some examples, to support communicating the additional one or more random access messages, the random access procedure communication element **1235** may be configured as or otherwise support a means for receiving a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.

(165) FIG. **13** shows a diagram of a system **1300** including a device **1305** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The device **1305** may be an example of or include the components of a device **1005**, a device **1105**, or a network entity **105** as described herein. The device **1305** may communicate with one or more network entities **105**, one or more UEs **115**, or any combination thereof, which may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device **1305** may include components that support outputting and obtaining communications, such as a communications manager **1320**, a

transceiver **1310**, an antenna **1315**, a memory **1325**, code **1330**, and a processor **1335**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **1340**).

(166) The transceiver **1310** may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver **1310** may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver **1310** may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the device **1305** may include one or more antennas **1315**, which may be capable of transmitting or receiving wireless transmissions (e.g., concurrently). The transceiver **1310** may also include a modem to modulate signals, to provide the modulated signals for transmission (e.g., by one or more antennas **1315**, by a wired transmitter), to receive modulated signals (e.g., from one or more antennas **1315**, from a wired receiver), and to demodulate signals. The transceiver **1310**, or the transceiver **1310** and one or more antennas **1315** or wired interfaces, where applicable, may be an example of a transmitter **1015**, a transmitter **1115**, a receiver **1010**, a receiver **1110**, or any combination thereof or component thereof, as described herein. In some examples, the transceiver may be operable to support communications via one or more communications links (e.g., a communication link **125**, a backhaul communication link **120**, a midhaul communication link **162**, a fronthaul communication link **168**).

(167) The memory **1325** may include RAM and ROM. The memory **1325** may store computer-readable, computer-executable code **1330** including instructions that, when executed by the processor **1335**, cause the device **1305** to perform various functions described herein. The code **1330** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **1330** may not be directly executable by the processor **1335** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **1325** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(168) The processor **1335** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, an ASIC, a CPU, a GPU, an FPGA, a microcontroller, a programmable logic device, discrete gate or transistor logic, a discrete hardware component, or any combination thereof). In some cases, the processor **1335** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **1335**. The processor **1335** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1325**) to cause the device **1305** to perform various functions (e.g., functions or tasks supporting bandwidth part configuration for reduced capability devices). For example, the device **1305** or a component of the device **1305** may include a processor **1335** and memory **1325** coupled with the processor **1335**, the processor **1335** and memory **1325** configured to perform various functions described herein. The processor **1335** may be an example of a cloud-computing platform (e.g., one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (e.g., by executing code **1330**) to perform the functions of the device **1305**.

(169) In some examples, a bus **1340** may support communications of (e.g., within) a protocol layer of a protocol stack. In some examples, a bus **1340** may support communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack), which may include communications performed within a component of the device **1305**, or between different components of the device **1305** that may be co-located or located in different locations (e.g., where the device **1305** may refer to a system in which one or more of the communications manager **1320**, the transceiver **1310**, the memory **1325**, the code **1330**, and the processor **1335** may be located in one of the different components or divided between different components).

(170) In some examples, the communications manager **1320** may manage aspects of communications with a core network **130** (e.g., via one or more wired or wireless backhaul links). For example, the communications manager **1320** may manage the transfer of data communications for client devices, such as one or more UEs **115**. In some examples, the communications manager **1320** may manage communications with other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** in cooperation with other network entities **105**. In some examples, the communications manager **1320** may support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**.
(171) Additionally, or alternatively, the communications manager **1320** may support wireless communications at a network entity in accordance with examples as disclosed herein. For example, the communications manager **1320** may be configured as or otherwise support a means for transmitting a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The communications manager **1320** may be configured as or otherwise support a means for receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.
(172) By including or configuring the communications manager **1320** in accordance with examples as described herein, the device **1305** may support techniques for improved communication reliability, reduced latency, improved user experience related to reduced processing, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, improved utilization of processing capability, or any combination thereof.
(173) In some examples, the communications manager **1320** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **1310**, the one or more antennas **1315** (e.g., where applicable), or any combination thereof. Although the communications manager **1320** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1320** may be supported by or performed by the processor **1335**, the memory **1325**, the code **1330**, the transceiver **1310**, or any combination thereof. For example, the code **1330** may include instructions executable by the processor **1335** to cause the device **1305** to perform various aspects of bandwidth part configuration for reduced capability devices as described herein, or the processor **1335** and the memory **1325** may be otherwise configured to perform or support such operations.
(174) FIG. **14** shows a flowchart illustrating a method **1400** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The operations of the method **1400** may be implemented by a UE or its components as described herein. For example, the operations of the method **1400** may be performed by a UE **115** as described with reference to FIGS. **1** through **9**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.
(175) At **1405**, the method may include receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by an SSB reception

component **825** as described with reference to FIG. **8**.
(176) At **1410**, the method may include selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by a RO selection component **830** as described with reference to FIG. **8**.
(177) At **1415**, the method may include transmitting, as part of a random access procedure, the random access preamble within the random access occasion. The operations of **1415** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1415** may be performed by a random access preamble transmission component **835** as described with reference to FIG. **8**.
(178) FIG. **15** shows a flowchart illustrating a method **1500** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The operations of the method **1500** may be implemented by a UE or its components as described herein. For example, the operations of the method **1500** may be performed by a UE **115** as described with reference to FIGS. **1** through **9**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.
(179) At **1505**, the method may include receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The operations of **1505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1505** may be performed by an SSB reception component **825** as described with reference to FIG. **8**.
(180) At **1510**, the method may include receiving an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the set of multiple random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts. The operations of **1510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1510** may be performed by an SSB reception component **825** as described with reference to FIG. **8**.
(181) At **1515**, the method may include selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The operations of **1515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1515** may be performed by a RO selection component **830** as described with reference to FIG. **8**.
(182) At **1520**, the method may include transmitting, as part of a random access procedure, the random access preamble within the random access occasion. The operations of **1520** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1520** may be performed by a random access preamble transmission component **835** as described with reference to FIG. **8**.
(183) FIG. **16** shows a flowchart illustrating a method **1600** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The operations of the method **1600** may be implemented by a UE or its components as described herein. For example, the operations of the method **1600** may be performed by a UE **115** as described with reference to FIGS. **1** through **9**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-

purpose hardware.

(184) At **1605**, the method may include receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part, wherein at most one of the set of multiple the random access occasions being within the first initial uplink bandwidth part, a remainder of the set of multiple the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part. The operations of **1605** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1605** may be performed by an SSB reception component **825** as described with reference to FIG. **8**.

(185) At **1610**, the method may include selecting a random access occasion of the set of multiple random access occasions for transmission of a random access preamble from the UE to the network entity. The operations of **1610** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1610** may be performed by a RO selection component **830** as described with reference to FIG. **8**.

(186) At **1615**, the method may include transmitting, as part of a random access procedure, the random access preamble within the random access occasion. The operations of **1615** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1615** may be performed by a random access preamble transmission component **835** as described with reference to FIG. **8**.

(187) FIG. **17** shows a flowchart illustrating a method **1700** that supports bandwidth part configuration for reduced capability devices in accordance with one or more aspects of the present disclosure. The operations of the method **1700** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1700** may be performed by a network entity as described with reference to FIGS. **1** through **5** and **10** through **13**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

(188) At **1705**, the method may include transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a set of multiple random access occasions, at least one of the set of multiple random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part. The operations of **1705** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1705** may be performed by an SSB transmission element **1225** as described with reference to FIG. **12**.

(189) At **1710**, the method may include receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions. The operations of **1710** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1710** may be performed by a random access preamble reception element **1230** as described with reference to FIG. **12**.

(190) The following provides an overview of aspects of the present disclosure:

(191) Aspect 1: A method for wireless communications at a UE, comprising: receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; selecting a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity; and transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

(192) Aspect 2: The method of aspect 1, wherein receiving the message further comprises: receiving an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the plurality of random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.
(193) Aspect 3: The method of aspect 2, further comprising: communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble is transmitted.
(194) Aspect 4: The method of any of aspects 2 through 3, further comprising: determining that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts; incrementing a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble and on whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble; and retransmitting the random access preamble in accordance with the preamble power ramping parameter.
(195) Aspect 5: The method of any of aspects 2 through 4, further comprising: determining that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts; incrementing a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble, regardless of whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble; and retransmitting the random access preamble in accordance with the preamble power ramping parameter.
(196) Aspect 6: The method of any of aspects 2 through 5, further comprising: determining that transmission of the random access preamble failed; and retransmitting the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, wherein the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.
(197) Aspect 7: The method of any of aspects 2 through 6, further comprising: determining that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts; and retransmitting the random access preamble via the first attempted initial uplink bandwidth part, wherein the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions per initial uplink bandwidth part.
(198) Aspect 8: The method of any of aspects 2 through 7, wherein selecting the random access occasion further comprises: monitoring an initial downlink bandwidth part for paging occasions during the random access procedure; identifying, from the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, a corresponding initial uplink bandwidth part that corresponds to the initial downlink bandwidth part; and selecting the random access occasion that is within the corresponding initial uplink bandwidth part based at least in part on the corresponding initial uplink bandwidth part corresponding to the initial downlink bandwidth part.
(199) Aspect 9: The method of any of aspects 1 through 8, wherein at most one of the plurality of the random access occasions is within the first initial uplink bandwidth part, a remainder of the plurality of the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.

(200) Aspect 10: The method of aspect 9, further comprising: receiving information that is indicative of at least one of an offset parameter or a binary direction parameter, wherein the binary direction parameter indicates a direction in which the offset parameter is to be applied with respect to a baseline resource block; determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part based at least in part on the information.
(201) Aspect 11: The method of aspect 10, wherein receiving the information that is indicative of at least one of the offset parameter or the binary direction parameter further comprises: receiving indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis.
(202) Aspect 12: The method of any of aspects 10 through 11, wherein determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part further comprises: determining, based at least in part on the offset parameter, the binary direction parameter, the baseline resource block, and a first frequency resource associated with a first random access occasion of the plurality of random access occasions; and determining additional frequency resources associated with other random access occasions of the plurality of random access occasions, the additional frequency resources determined in relation to the first frequency resource.
(203) Aspect 13: The method of any of aspects 1 through 12, further comprising: communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.
(204) Aspect 14: The method of aspect 13, wherein communicating the additional one or more random access messages further comprises: receiving a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(205) Aspect 15: The method of any of aspects 13 through 14, wherein communicating the additional one or more random access messages further comprises: transmitting a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(206) Aspect 16: The method of any of aspects 13 through 15, wherein communicating the additional one or more random access messages further comprises: transmitting a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.
(207) Aspect 17: The method of any of aspects 1 through 16, wherein the message is received through a system information block, a synchronization signal block, downlink control information, a medium access control control element, radio resource control signaling, a handover command, or any combination thereof.
(208) Aspect 18: A method for wireless communications at a network entity, comprising: transmitting a message that indicates a first initial uplink bandwidth part for use by a UE in communications with a network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; and receiving, as part of a random access procedure, the random access preamble within the random access occasion.
(209) Aspect 19: The method of aspect 18, wherein receiving the message further comprises: transmitting an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the plurality of random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.
(210) Aspect 20: The method of aspect 19, further comprising: communicating additional one or

more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts in which the random access preamble is transmitted.
(211) Aspect 21: The method of any of aspects 19 through 20, further comprising: determining that transmission of the random access preamble failed; and receiving a retransmission of the random access preamble via any of the first initial uplink bandwidth part or the additional initial uplink bandwidth parts, wherein the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the additional initial uplink bandwidth parts, collectively.
(212) Aspect 22: The method of any of aspects 18 through 21, wherein at most one of the plurality of the random access occasions is within the first initial uplink bandwidth part, a remainder of the plurality of the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.
(213) Aspect 23: The method of aspect 22, further comprising: transmitting information that is indicative of at least one of an offset parameter or a binary direction parameter, wherein the binary direction parameter indicates a direction in which the offset parameter is to be applied with respect to a baseline resource block; and determining one or more of the frequency resources that are outside of the first initial uplink bandwidth part based at least in part on the information.
(214) Aspect 24: The method of aspect 23, wherein transmitting the information that is indicative of at least one of the offset parameter or the binary direction parameter further comprises: transmitting indications of at least one of the offset parameter or the binary direction parameter on a per-random access occasion basis.
(215) Aspect 25: The method of any of aspects 18 through 24, further comprising: communicating additional one or more random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.
(216) Aspect 26: The method of aspect 25, wherein communicating the additional one or more random access messages further comprises: transmitting a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(217) Aspect 27: The method of any of aspects 25 through 26, wherein communicating the additional one or more random access messages further comprises: receiving a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
(218) Aspect 28: The method of any of aspects 25 through 27, wherein communicating the additional one or more random access messages further comprises: receiving a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.
(219) Aspect 29: An apparatus for wireless communications at a UE, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 1 through 17.
(220) Aspect 30: An apparatus for wireless communications at a UE, comprising at least one means for performing a method of any of aspects 1 through 17.
(221) Aspect 31: A non-transitory computer-readable medium storing code for wireless communications at a UE, the code comprising instructions executable by a processor to perform a method of any of aspects 1 through 17.
(222) Aspect 32: An apparatus for wireless communications at a network entity, comprising a

processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 18 through 28.

(223) Aspect 33: An apparatus for wireless communications at a network entity, comprising at least one means for performing a method of any of aspects 18 through 28.

(224) Aspect 34: A non-transitory computer-readable medium storing code for wireless communications at a network entity, the code comprising instructions executable by a processor to perform a method of any of aspects 18 through 28.

(225) It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.

(226) Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies, including future systems and radio technologies, not explicitly mentioned herein.

(227) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(228) The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, a CPU, a GPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(229) The functions described herein may be implemented in hardware, software executed by a processor, or any combination thereof. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, or functions, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(230) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation,

non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, phase change memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.
(231) As used herein, including in the claims, "or" as used in a list of items (e.g., a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase "based on" shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as "based on condition A" may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase "based on" shall be construed in the same manner as the phrase "based at least in part on." As used herein, the term "and/or," when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination
(232) The term "determine" or "determining" encompasses a variety of actions and, therefore, "determining" can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database or another data structure), ascertaining and the like. Also, "determining" can include receiving (such as receiving information), accessing (such as accessing data in a memory) and the like. Also, "determining" can include resolving, obtaining, selecting, choosing, establishing and other such similar actions.
(233) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.
(234) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term "example" used herein means "serving as an example, instance, or illustration," and not "preferred" or "advantageous over other examples." The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.
(235) The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations

without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus for wireless communications at a user equipment (UE), comprising: at least one processor; and memory coupled to the at least one processor, the memory storing instructions executable by the at least one processor to cause the UE to: receive a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; select a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity; and transmit, as part of a random access procedure, the random access preamble within the random access occasion.
2. The apparatus of claim 1, wherein the instructions to receive the message are further executable by the at least one processor to cause the UE to: receive an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the plurality of random access occasions being in a respective one of the first initial uplink bandwidth part or additional initial uplink bandwidth parts.
3. The apparatus of claim 2, wherein the instructions are further executable by the at least one processor to cause the UE to: communicate one or more additional random access messages that are associated with transmission of the random access preamble, the one or more additional random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the one or more additional initial uplink bandwidth parts in which the random access preamble is transmitted.
4. The apparatus of claim 2, wherein the instructions are further executable by the at least one processor to cause the UE to: determine that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts; increment a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble and on whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble; and retransmit the random access preamble in accordance with the preamble power ramping parameter.
5. The apparatus of claim 2, wherein the instructions are further executable by the at least one processor to cause the UE to: determine that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts; increment a preamble power ramping parameter based at least in part on failure of the transmission of the random access preamble, regardless of whether the first attempted initial uplink bandwidth part is to be reused for retransmission of the random access preamble; and retransmit the random access preamble in accordance with the preamble power ramping parameter.
6. The apparatus of claim 2, wherein the instructions are further executable by the at least one processor to cause the UE to: determine that transmission of the random access preamble failed; and retransmit the random access preamble via any of the first initial uplink bandwidth part or the one or more additional initial uplink bandwidth parts, wherein the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts,

collectively.
7. The apparatus of claim 2, wherein the instructions are further executable by the at least one processor to cause the UE to: determine that transmission of the random access preamble failed, wherein the random access preamble was transmitted within a first attempted initial uplink bandwidth part selected from the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts; and retransmit the random access preamble via the first attempted initial uplink bandwidth part, wherein the retransmitting is in accordance with a maximum number of retransmissions parameter that applies to retransmissions per initial uplink bandwidth part.
8. The apparatus of claim 2, wherein the instructions to select the random access occasion are further executable by the at least one processor to cause the UE to: monitor an initial downlink bandwidth part for paging occasions during the random access procedure; identify, from the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts, a corresponding initial uplink bandwidth part that corresponds to the initial downlink bandwidth part; and select the random access occasion that is within the corresponding initial uplink bandwidth part based at least in part on the corresponding initial uplink bandwidth part corresponding to the initial downlink bandwidth part.
9. The apparatus of claim 1, wherein at most one of the plurality of the random access occasions is within the first initial uplink bandwidth part, a remainder of the plurality of the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.
10. The apparatus of claim 9, wherein the instructions are further executable by the at least one processor to cause the UE to: receive information that is indicative of at least one of a random access occasion offset parameter or a random access occasion binary direction parameter, wherein the random access occasion binary direction parameter indicates a direction in which the random access occasion offset parameter is to be applied with respect to a baseline resource block; and determine one or more of the frequency resources that are outside of the first initial uplink bandwidth part based at least in part on the information.
11. The apparatus of claim 10, wherein the instructions to receive the information that is indicative of at least one of the random access occasion offset parameter or the random access occasion binary direction parameter are further executable by the at least one processor to cause the UE to: receive indications of at least one of the random access occasion offset parameter or the random access occasion binary direction parameter on a per-random access occasion basis.
12. The apparatus of claim 10, wherein the instructions to determine one or more of the frequency resources that are outside of the first initial uplink bandwidth part are further executable by the at least one processor to cause the UE to: determine, based at least in part on the random access occasion offset parameter, the random access occasion binary direction parameter, the baseline resource block, and a first frequency resource associated with a first random access occasion of the plurality of random access occasions; and determine additional frequency resources associated with other random access occasions of the plurality of random access occasions, the additional frequency resources determined in relation to the first frequency resource.
13. The apparatus of claim 1, wherein the instructions are further executable by the at least one processor to cause the UE to: communicate one or more additional random access messages that are associated with transmission of the random access preamble, the one or more additional random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.
14. The apparatus of claim 13, wherein the instructions to communicate the one or more additional random access messages are further executable by the at least one processor to cause the UE to: receive a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.
15. The apparatus of claim 13, wherein the instructions to communicate the one or more additional

random access messages are further executable by the at least one processor to cause the UE to: transmit a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

16. The apparatus of claim 13, wherein the instructions to communicate the one or more additional random access messages are further executable by the at least one processor to cause the UE to: transmit a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.

17. The apparatus of claim 1, wherein the message is received through a system information block, a synchronization signal block, downlink control information, a medium access control control element, radio resource control signaling, a handover command, or any combination thereof.

18. An apparatus for wireless communications at a network entity, comprising: at least one processor; and memory coupled to the at least one processor, the memory storing instructions executable by the at least one processor to cause the network entity to: transmit a message that indicates a first initial uplink bandwidth part for use by a user equipment (UE) in communications with the network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; and receive, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.

19. The apparatus of claim 18, wherein the instructions to transmit the message are further executable by the at least one processor to cause the network entity to: transmit an indication of one or more additional initial uplink bandwidth parts for use by the UE in the communications with the network entity, each of the plurality of random access occasions being in a respective one of the first initial uplink bandwidth part or one or more additional initial uplink bandwidth parts.

20. The apparatus of claim 19, wherein the instructions are further executable by the at least one processor to cause the network entity to: communicate one or more additional random access messages that are associated with transmission of the random access preamble, the one or more random access messages being communicated in frequency resources that are within a same one of the first initial uplink bandwidth part or the one or more additional initial uplink bandwidth parts in which the random access preamble is transmitted.

21. The apparatus of claim 19, wherein the instructions are further executable by the at least one processor to cause the network entity to: determine that transmission of the random access preamble failed; and receive a retransmission of the random access preamble via any of the first initial uplink bandwidth part or the one or more additional initial uplink bandwidth parts, wherein the retransmission is in accordance with a maximum number of retransmissions parameter that applies to retransmissions across all of the first initial uplink bandwidth part and the one or more additional initial uplink bandwidth parts, collectively.

22. The apparatus of claim 18, wherein at most one of the plurality of the random access occasions is within the first initial uplink bandwidth part, a remainder of the plurality of the random access occasions being outside of the first initial uplink bandwidth part but still associated with the first initial uplink bandwidth part.

23. The apparatus of claim 22, wherein the instructions are further executable by the at least one processor to cause the network entity to: transmit information that is indicative of at least one of a random access occasion offset parameter or a random access occasion binary direction parameter, wherein the random access occasion binary direction parameter indicates a direction in which the random access occasion offset parameter is to be applied with respect to a baseline resource block; and determine one or more of the frequency resources that are outside of the first initial uplink bandwidth part based at least in part on the information.

24. The apparatus of claim 23, wherein the instructions to transmit the information that is indicative

of at least one of the random access occasion offset parameter or the random access occasion binary direction parameter are further executable by the at least one processor to cause the network entity to: transmit indications of at least one of the random access occasion offset parameter or the random access occasion binary direction parameter on a per-random access occasion basis.

25. The apparatus of claim 18, wherein the instructions are further executable by the at least one processor to cause the network entity to: communicate one or more additional random access messages that are associated with transmission of the random access preamble, the one or more additional random access messages being communicated in frequency resources that are within a selected initial uplink bandwidth part that corresponds to the random access occasion.

26. The apparatus of claim 25, wherein the instructions to communicate the one or more additional random access messages are further executable by the at least one processor to cause the network entity to: transmit a random access response message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

27. The apparatus of claim 25, wherein the instructions to communicate the one or more additional random access messages are further executable by the at least one processor to cause the network entity to: receive a connection request message separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap.

28. The apparatus of claim 25, wherein the instructions to communicate the one or more additional random access messages are further executable by the at least one processor to cause the network entity to: receive a random access request payload separated in time from transmission of the random access preamble by a duration of time that includes a bandwidth part switching time gap, the random access procedure being a two-step random access procedure.

29. A method for wireless communications at a user equipment (UE), comprising: receiving a message that indicates a first initial uplink bandwidth part for use by the UE in communications with a network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; selecting a random access occasion of the plurality of random access occasions for transmission of a random access preamble from the UE to the network entity; and transmitting, as part of a random access procedure, the random access preamble within the random access occasion.

30. A method for wireless communications at a network entity, comprising: transmitting a message that indicates a first initial uplink bandwidth part for use by a user equipment (UE) in communications with the network entity, the message also indicating a plurality of random access occasions, at least one of the plurality of random access occasions being in frequency resources that are outside of the first initial uplink bandwidth part; and receiving, as part of a random access procedure, a random access preamble within a random access occasion selected from the plurality of random access occasions.